



Engineering
Portfolio

DRIVEN BY VOLTS, DEFINED BY VISION

VOLT.EDGE



National
Robotics League

India's Biggest Robotics Championship
for Middle & High School Students



Team VoltEdge

Representing at NRL 2025

Team ID: 007



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Team Introduction

Team Story And Vision

VoltEdge began with a moment that wasn't planned — a message sent on the night of 4 October, just before our midterm exams. Omer reached out to Abdullah with the details of the National Robotics League, and within minutes, the idea of forming a team took shape. None of us had a robot, a workspace, or a plan... but we had curiosity. And sometimes, that is enough to begin.

By 9 October, a day before exams, we registered as Team VoltEdge — Team ID 007, trusting that we would figure everything else out along the way. When our BaseBot kit finally arrived in Bhatkal on 20 October, the spark turned into commitment. What followed from 22 to 27 October was six days of continuous building — tightening screws, aligning plates, wiring circuits, testing motors, fixing mistakes, and learning at every step. Those six days shaped our identity more than anything else: a team that learns fast, adapts fast, and moves forward no matter how busy life gets.

Our vision grew from these moments.

VoltEdge was never just about building a robot. It was about proving that a group of students from a small coastal town could enter a national arena with confidence. It was about showing younger students that engineering isn't far away — it's something they can touch, build, and experience. It was about finding a place where each of us could use our strengths: electronics, mechanics, programming, communication, design, and leadership.

Our vision is simple and powerful:

To build engineering talent in our town, to inspire the next generation through robotics, and to step into the NRL arena not only as competitors — but as a team defined by discipline, purpose, and unity.

VoltEdge believes in doing things with honesty, learning through mistakes, and pushing ourselves beyond what we thought possible. Whether it is building a robot in six days, running workshops during a packed college schedule, managing outreach, or preparing a portfolio days before the event — our story is built on action.

And as we submit this engineering portfolio on 30 November 2025, days before stepping into the IIT Bombay arena, our vision remains unchanged:

Build with purpose.

Compete with discipline.

Inspire with impact.

VoltEdge — Driven by Volts, Defined by Vision



Team Roles And Responsibilities

Every member of VoltEdge carries a role shaped by their strengths, their instincts, and the responsibilities they naturally grew into during the journey from 22 October to 27 October. As the robot took form piece by piece, each person became essential to a different part of its creation — some on the mechanical side, some on electronics, some on programming and documentation, and some on strategy and operations. Together, these roles formed the backbone of Team VoltEdge.

Abdullah Saami Sada serves as the team's Documentation Head and a Programming Member in the Pit Team. He is responsible for building the entire engineering portfolio, structuring team communication, maintaining clarity in technical decisions, and shaping the written identity of VoltEdge. During the build phase, Abdullah supported programming tasks, assisted in software setup, and ensured that documentation accurately reflected the team's work from day one.

Mohiddin Ahmed Motiya contributes as both a Human Player in the driver team and the Brand Manager of VoltEdge. On the field, he ensures smooth human-robot coordination, while off the field, he manages branding, identity, and the visual communication of the team. His decisions shape how VoltEdge is represented to the school community, online audiences, and future outreach programs.

Omer Ruknuddin, the Lead Programmer and Primary Driver, is responsible for the robot's software architecture, controller mapping, movement logic, and performance tuning. During matches, Omer takes control of the robot as the main driver, combining technical understanding with field awareness. His role bridges programming precision with real-time decision-making in the arena.

Mohammed Zaid Fareed leads the electronics domain as the Electronics Lead and a Pit Team Electronics Member. He manages wiring, circuit stability, power routing, sensor integration, and all electronic safety checks. His responsibility includes ensuring that every component remains functional under stress, making him the electrical backbone of the robot during both construction and testing.

Ahmed Irfan Akrami, the Mechanical Lead and a Pit Team Mechanical Member, is responsible for the robot's structural integrity. He oversees chassis alignment, arm and gripper assembly, torque balancing, wheel fitting, and all mechanical problem-solving. His precision ensures the robot's stability, strength, and consistency during matches and testing cycles.

Shamveel Bukhari Khateeb serves as the Driver Strategist, Videographer, and Visual Head. On-field, he supports Omer as a strategist, guiding movements, timing, and overall match flow. Off-field, he handles video production, documentation footage, and all visual storytelling for VoltEdge. His footage and media work capture the team's journey and enhance communication with the community and the NRL audience.

Mr. Nuhail Damudi is more than a mentor to Team VoltEdge — he is the person who first opened the door to robotics for us back when we were in 9th grade. What began as a simple introduction has grown into a journey that shaped our skills, our mindset, and ultimately the creation of Team VoltEdge itself. With a strong foundation in robotics and innovation, he guides us with clarity, patience, and a vision grounded in real engineering discipline. His mentorship pushes us to think deeper, design smarter, and approach every challenge with purpose. Under his guidance, VoltEdge evolved from curious beginners into a focused, confident team ready to compete on a national stage.

Together, these roles form the structure of Team VoltEdge.

Each responsibility interlocks with another, creating a cycle of support, precision, and teamwork that defines how VoltEdge builds, tests, documents, and prepares for the National Robotics League.

Team Photos and Work Approach

The visual story of VoltEdge is captured through a series of photos that reflect not just the robot we built, but the way we worked together. Our team's images show late-evening work sessions, the crowded tables filled with screws and tools, moments of collaboration between mechanical and electronics teams, and the quiet focus on screens during programming tests. These photos document our journey from receiving the kit in Bhatkal to assembling the robot step by step through structured teamwork, constant communication, and shared problem-solving.

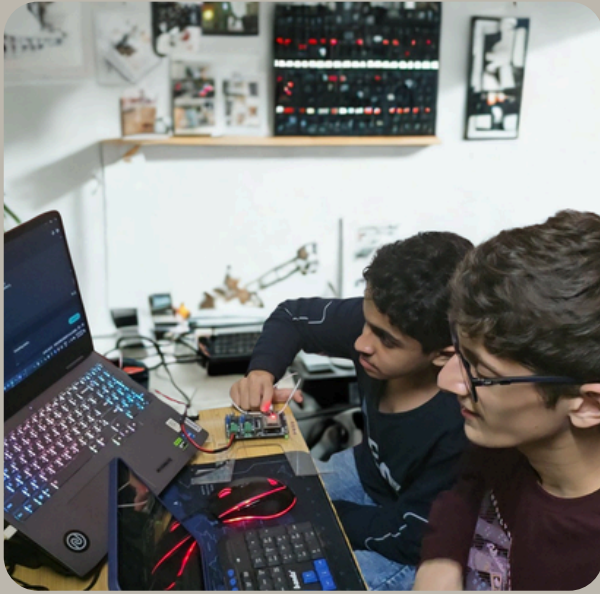


Our work approach is built around discipline and consistency. From 22 October until the robot was completed on 27 October, the team met regularly—often in the evenings after college hours—to divide tasks, review progress, and solve issues in real time. Mechanical and electronics members worked side by side, ensuring that chassis fitting, wiring, and component placement aligned perfectly. Programmers joined these sessions to test motor response and refine control logic as soon as components were installed. This cycle of “build → test → adjust” repeated continuously, allowing the robot to evolve smoothly across the five-day build window.

During these sessions, every member naturally stepped into their responsibility: Irfan handled the difficult mechanical alignments, Zaid managed the wiring and electronic stability, Omer tested movements and logic on the controller, Abdullah documented each stage while supporting software tasks, Shamveel captured the build through photos and videos, and Mohiddin assisted with coordination and brand presentation. The team's work style focused on precision, clarity, and mutual support—no task was ever done alone.



Our photos display this atmosphere: the concentration around the workbench, the scattered components slowly forming into a complete machine, the late-night energy, and the small celebrations after each successful test. This visual record not only shows the robot's construction but also represents how VoltEdge operates—as a coordinated unit where creativity, discipline, and teamwork merge into one story.



Prototype And Basebot Assembly

Overview Of Basebot Design And Assembly

The BaseBot became the starting point of VoltEdge's entire engineering journey. When the kit arrived in Bhatkal on 20 October, it was nothing more than a collection of acrylic plates, motors, screws, spacers, sensors, and electronic modules — but together, these parts formed a foundation strong enough for a national-level robot. We began assembly on 22 October, knowing that every component we placed would determine how the robot behaved in testing and how reliably it would perform in the NRL arena.

The BaseBot's design follows a simple yet effective architecture: a dual-layer acrylic chassis supported by steel and aluminium brackets, a four-motor differential drive system for movement, and a servo-driven arm and gripper for collecting charges. Each piece has a purpose — the acrylic plates give structure, the brackets ensure motor alignment, the spacers maintain balance, and the electronics mounted on top bring the entire system to life.

During the assembly days from 22 to 27 October, we slowly watched the robot transition from scattered parts into a complete prototype. Irfan aligned the chassis plates and fitted the mechanical pieces, Zaid connected the electronics and routed the wiring cleanly across the robot, Omer and Abdullah tested the initial motor responses through the ESP32, and the rest of the team assisted wherever an extra pair of hands was needed. As each section was completed — the drivetrain, the gripper, the arm linkages, the control hub — the robot began to move, respond, and function the way it was meant to.

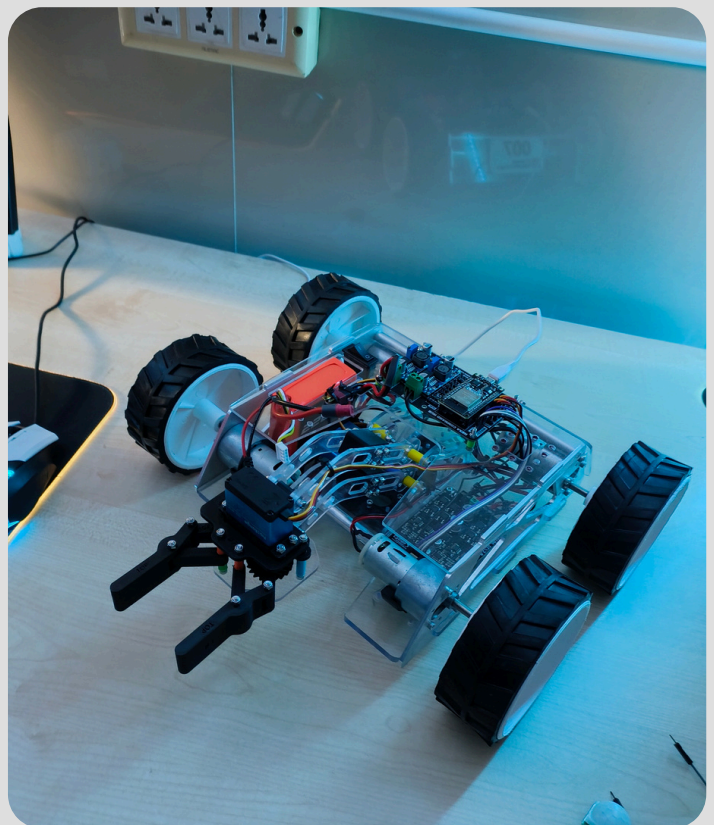
By 27 October, the BaseBot was fully assembled and operational. The design gave us a reliable platform to test, learn, and prepare for modifications. These early days taught us how each subsystem interacted with the others, setting the stage for tuning, autonomous work, and match strategy in the weeks that followed.

Testing And Early Prototype Results

Once the BaseBot assembly was completed on 27 October, the robot entered its most important phase: testing. These early tests were where VoltEdge first understood how the robot behaved beyond the instruction manual. The initial drive test showed us small imperfections — a slight pull toward one side, uneven wheel grip, and minor vibration in the chassis. These weren't failures; they were signals telling us where adjustments were needed. Each test revealed something new about how the robot responded to turns, stops, and quick directional changes.

As we continued testing through the evening sessions, we refined the robot step by step. The drivetrain was inspected repeatedly to correct alignment, the arm's movement checked for smooth lifting, and wiring was tightened where vibration caused loose connections. We tested the motor driver response, ESP32 communication, gripper precision, and wheel stability on different surfaces. With every cycle of “test → observe → fix,” the robot became more predictable and more stable.

These early prototype sessions taught us how the robot behaved under real motion. They showed us which parts needed reinforcement, which movements needed refining, and how the robot transitioned from a newly assembled structure into a functional machine. By the end of our testing phase, the BaseBot moved cleanly, lifted consistently, and responded accurately to controller inputs — giving VoltEdge the confidence to progress into strategy planning, programming refinement, and autonomous preparation.



Meet Our Robot

What Is Our Robot? Main Features And Capabilities

EdgeBot is Team VoltEdge's re-engineered evolution of the standard NRL BaseBot — a machine built not just to move, but to express our engineering character. From the moment we began designing, EdgeBot was imagined as a robot with presence: strong, stable, intelligent, and unmistakably ours. Its structure is wrapped in a custom 3D-printed shell inspired by monster-truck geometry, giving the bot a powerful identity and improved physical rigidity. The signature rear aero wing adds balance and control, working alongside the under-chassis downlights that illuminate the driving surface and provide tactical visibility during testing and matches.

But EdgeBot's appearance is only the beginning. Internally, the robot carries a set of intelligent sensor systems that elevate its capability far beyond the BaseBot. A gyroscope governs its turning accuracy, maintaining stability and preventing drift during sharp maneuvers. The ultrasonic module at the front gives EdgeBot the ability to detect charges and obstacles even before the driver reacts, allowing for a smoother approach and safer navigation.

For autonomous runs, a set of precisely mounted line-following sensors enable the robot to track paths reliably and execute planned routines with steady precision. Together, these elements make EdgeBot a machine that blends aesthetic impact, mechanical confidence, and smart sensing into a single cohesive system — an authentic VoltEdge engineering product.



Differences From The BaseBot

Although EdgeBot began its life as a BaseBot, nearly every major aspect has been transformed to suit our competitive needs. The most visible difference lies in its physical design. The BaseBot's plain open structure has been replaced with a fully custom 3D-printed monster-truck style body, giving EdgeBot a distinct and robust exterior.

This design strengthens the chassis while allowing for personalized mounting points, cleaner wiring, and better sensor placement. The rear aero wing, absent in the BaseBot, is a notable addition. It stabilizes the bot during acceleration and directional changes, enhancing grip and reducing unwanted lift. Additionally, the improved suspension system offers smoother handling on uneven surfaces. The integration of advanced telemetry systems further aids in real-time performance monitoring.

Internally, the BaseBot's simple electronic layout has been upgraded to a more refined system. The integration of a gyro sensor brings a new level of turning control that the BaseBot simply doesn't possess. The ultrasonic sensor, mounted at the front, adds an intelligent detection capability, enabling the robot to "sense" charges or obstacles — something beyond the BaseBot's default hardware. The autonomous functionality has also evolved through the addition of line-following sensors, giving EdgeBot the ability to perform guided automatic movements, unlike the BaseBot's purely driver-dependent operation.

Impact Of Modifications On Performance

Every modification we made to EdgeBot influences its performance in a meaningful way. The structural upgrades, especially the 3D-printed shell and rear wing, immediately improved stability. The bot now handles sharp turns with less body sway and maintains traction more effectively during rapid acceleration. The wing distributes weight more evenly, keeping the rear end grounded — a noticeable advantage when the bot needs to change direction quickly inside the arena.

The sensing system brought an even greater transformation. The gyroscope has had a major impact on accuracy by reducing drift and allowing our driver to trust the robot's directional control. Even minor turns feel more predictable and controlled. The ultrasonic sensor expanded the robot's awareness, enabling faster and safer charge detection. This not only improves the driver's confidence but also protects the robot from collisions and misalignment during matches.

Autonomous performance also saw a clear improvement. The BaseBot, with no inherent autonomous capabilities, relied heavily on manual control. EdgeBot's line-following sensors now allow it to track lines and execute planned paths with consistency. This transformed the robot's early-game strategy and opened opportunities for autonomous scoring that the BaseBot simply couldn't achieve.

Even the smaller changes made a noticeable difference. Downlights, while visually appealing, doubled as a functional diagnostic tool, letting us observe wheel movement and ground contact while testing. Cleaner wiring and refined controller mapping improved reliability and ease of maintenance, especially during pit operations. Altogether, these modifications created a robot that is faster, more stable, more aware, and far more competition-ready than the BaseBot ever was.

Mechanical Details

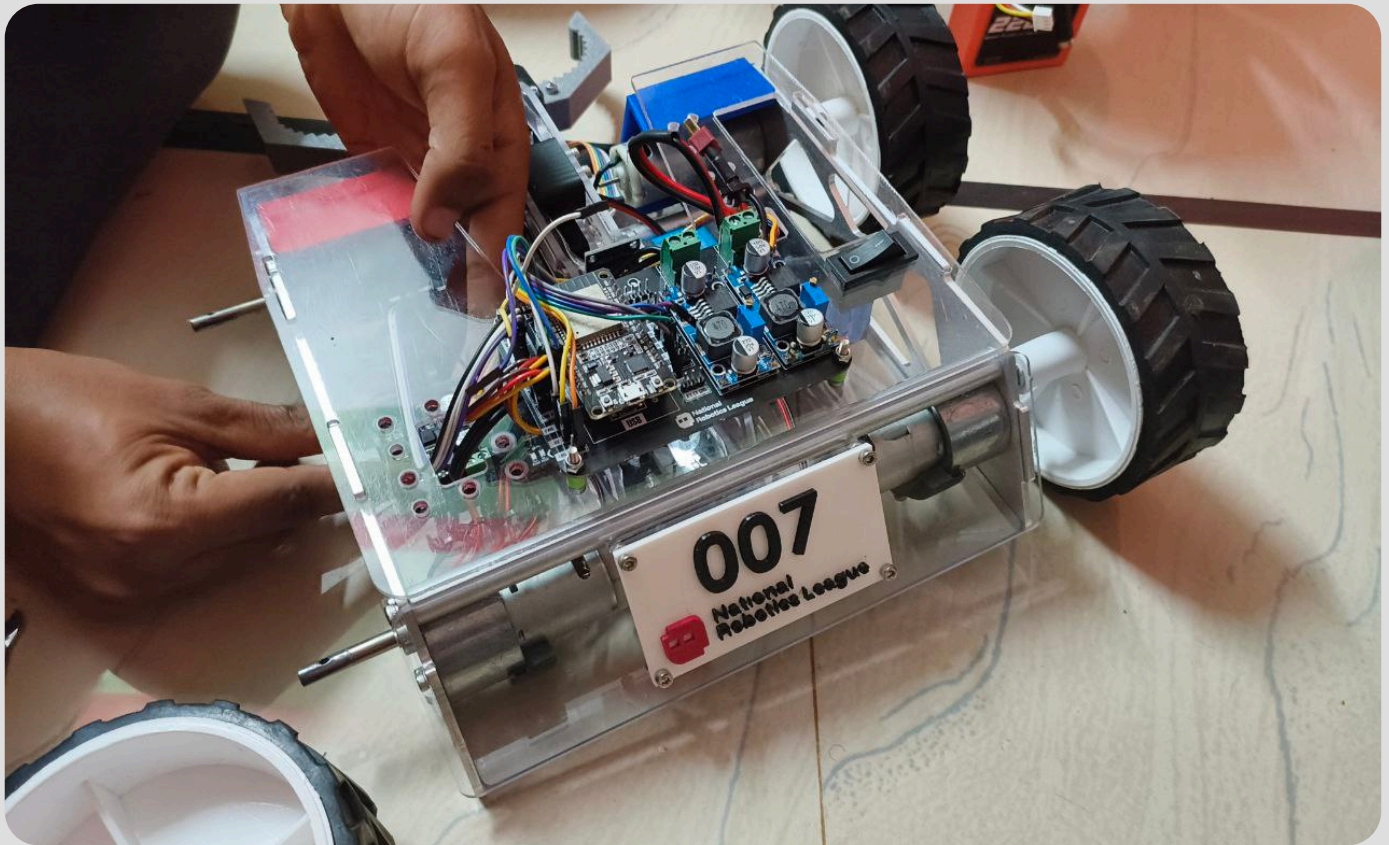
Basebot Chassis Design And Structure

The BaseBot's chassis became the structural backbone of our entire robot — the frame that supported every movement, every lift, and every test that followed. Built using dual-layer acrylic plates, the chassis provided a lightweight yet rigid foundation that made assembly straightforward and movement steady. The top and bottom plates were precisely cut to hold components like the motor driver, battery, ESP32 control hub, and servo mount, while still leaving enough spacing for clean wiring and airflow. These acrylic plates were supported by steel and aluminium side brackets, which ensured that the motors remained perfectly aligned even during fast turns and sudden directional changes.

As we assembled the chassis from 22 to 27 October, we began to understand why the BaseBot design uses this combination of materials. Acrylic offered smooth surfaces, clean fitting, and enough stiffness for the arm and gripper to operate without flex. The steel motor brackets carried the weight and torque of the drivetrain, preventing bending and protecting the motors from shifting under load. Together, they created a stable platform that kept the robot balanced during practice runs.

The final chassis measured 36 cm in width and 26 cm in length, extending to 37 cm when the gripper was fully forward. Its height ranged between 15 to 25 cm, depending on the arm's position. These dimensions gave the robot a compact and manoeuvrable profile, making it easier to navigate tight paths during testing and match strategy planning. Every hole, spacer, and bracket position was placed with purpose — supporting components in a way that made maintenance simple and movement reliable.

By the end of assembly, the chassis was not just a frame; it was the solid foundation that allowed every mechanical, electrical, and programming decision to work smoothly. It held the robot together physically and set the stage for all the improvements that VoltEdge would make in the days ahead.



Modified Mechanical Components

While the BaseBot provided a strong foundation, our mechanical work from 22 to 27 October showed us exactly where small adjustments could improve performance and stability. Under the guidance of our mechanical lead, Ahmed Irfan, we refined several parts of the structure to make the robot smoother, stronger, and more reliable during testing. The chassis plates were re-aligned to reduce minor flexing, ensuring the arm and gripper operated without strain. Wheel mounts were tightened and corrected to prevent uneven traction, helping the robot maintain a straight path during early drive tests. Certain sections of the frame were reinforced with properly balanced torque on screws and spacers, allowing the robot to remain stable even when the arm was lifted forward. These modifications were not redesigns — they were careful corrections and improvements that made the fully assembled BaseBot feel more controlled, predictable, and competition-ready during practice.

Electronic Details

BaseBot Wiring Diagrams And Setup

The wiring of the BaseBot formed the core electrical system that allowed the robot to move, lift, and respond the moment the controller connected. During the build sessions from 22 to 27 October, our electronics lead, Mohammed Zaid, laid out each component with careful planning before making the first connection. The MDD10 motor driver was mounted on the bottom acrylic plate, positioned so the motor inputs and power terminals were easy to reach. From there, each of the four DC motors was wired to its respective output channel, ensuring correct polarity so the drivetrain behaved predictably during early testing.

Above it, the HEXA Command Hub ESP32 acted as the robot's brain, sending PWM and direction signals through color-coded jumper wires that Zaid routed cleanly across the chassis. The servo for the arm was connected directly to its control pin on the ESP32, with enough wire slack to allow smooth vertical movement without strain. The power switch and battery holder were mounted firmly so the robot could be safely turned on and off without disturbing any internal wiring. Throughout the setup, cables were kept clear of the wheels and linkages to prevent tangling or accidental damage during rotation.

Every connection was double-checked before the first power-on: motor directions, servo response, battery terminals, and jumper stability. This careful wiring layout became the electrical backbone of EdgeBot — clean, organized, safe, and reliable during every drive test that followed.

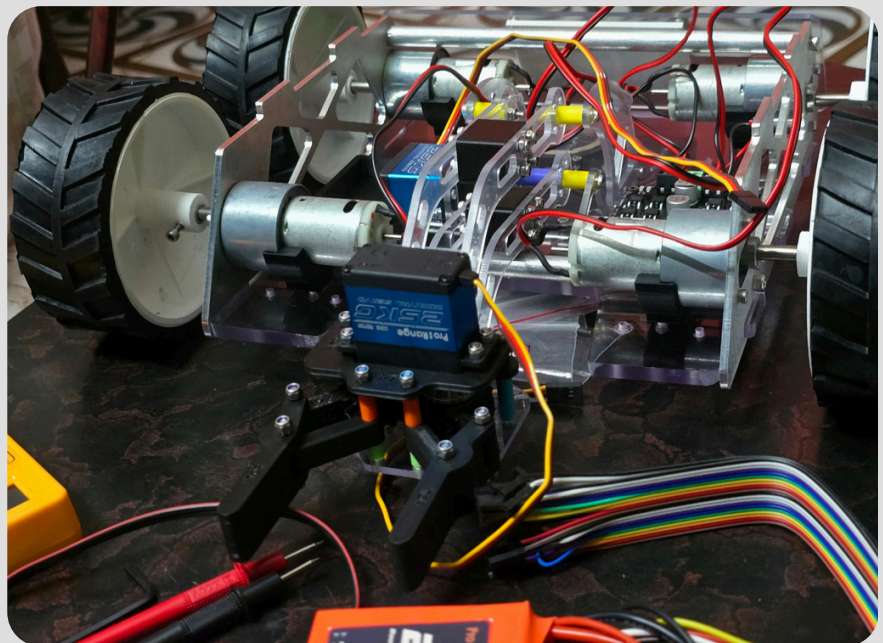
Some minerals in rocks can carry electricity, and
Certain metals found in rocks are used to make wires and
tiny parts inside circuits.

Modifications To Electrical System

As we began testing the BaseBot after the initial assembly, a few electrical improvements became necessary to make the robot more stable and easier to maintain. These were not major redesigns, but practical refinements led by our electronics lead, Mohammed Zaid, to ensure the wiring remained clean, safe, and consistent during repeated testing. The first adjustments focused on reorganizing cable routes so that none of the wires came close to the wheels, arm joints, or the rotating gripper area. This reduced strain on connectors and prevented loose wires from interfering with movement.

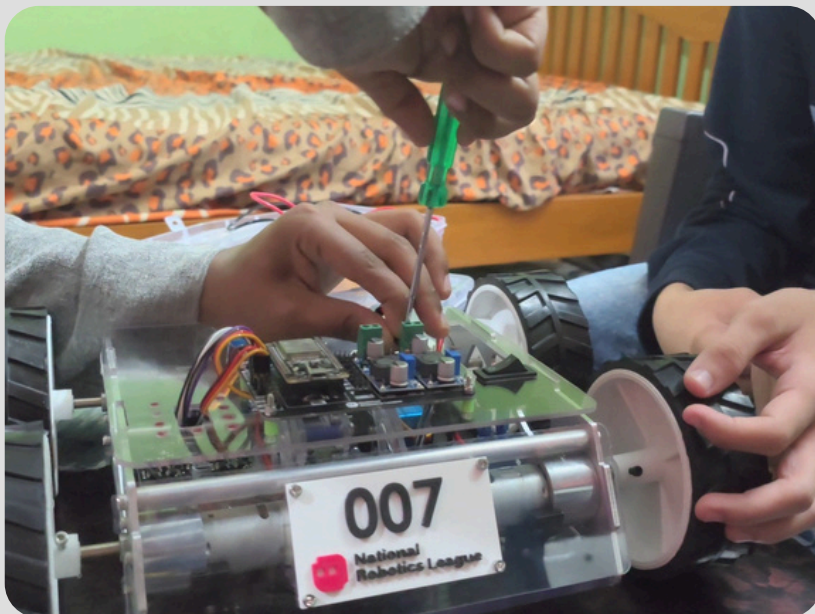
We also refined wire lengths, replacing or rerouting cables that were either too tight or too long, ensuring smoother internal spacing between the top and bottom chassis plates. The motor driver placement was slightly adjusted for better airflow and easier access to terminals during testing. Servo wires were also positioned more carefully to prevent tension when the arm lifted forward. These modifications didn't change the BaseBot's original electrical design — they simply made it more reliable, easier to troubleshoot, and safer for continuous operation.

Through these small but important improvements, the robot's electronic system became more organized and durable, giving VoltEdge the confidence needed for extended testing sessions and match preparation.



Safety Measures And Rewiring Explained

Safety was a constant priority throughout our build from 22 to 27 October, especially as we worked with exposed circuits, rotating motors, and a compact chassis layout. Before powering the robot for the first time, the team—led by our electronics head, Mohammed Zaid—reviewed every connection carefully to make sure no wire crossed moving parts, no terminal was loose, and no connector carried unnecessary tension. We avoided overtightening screws near the motor driver and ESP32 to prevent cracks in the acrylic base or damage to pins. The battery holder was secured firmly so that the robot could be turned and lifted during tests without shifting the power supply.



Whenever a rewiring task was needed, we followed one strict rule: power off before touching anything. Only after disconnecting the battery did we adjust motor wires, re-route jumper cables, or correct servo placement.

During early drive tests, some wires loosened due to vibration, and a few jumpers needed rearrangement for better spacing. Each time, the rewiring was done methodically—checking polarity twice, confirming motor orientation, and ensuring that the new routing kept the cables away from wheels and arm linkages. This careful approach prevented accidental short circuits, protected the components from stress, and kept the robot safe during repeated test runs.

By the end of the assembly phase, these safety habits became second nature to VoltEdge. Clean wiring, controlled rewiring, and constant checks ensured that EdgeBot remained stable and reliable, ready for both autonomous testing and match preparation.

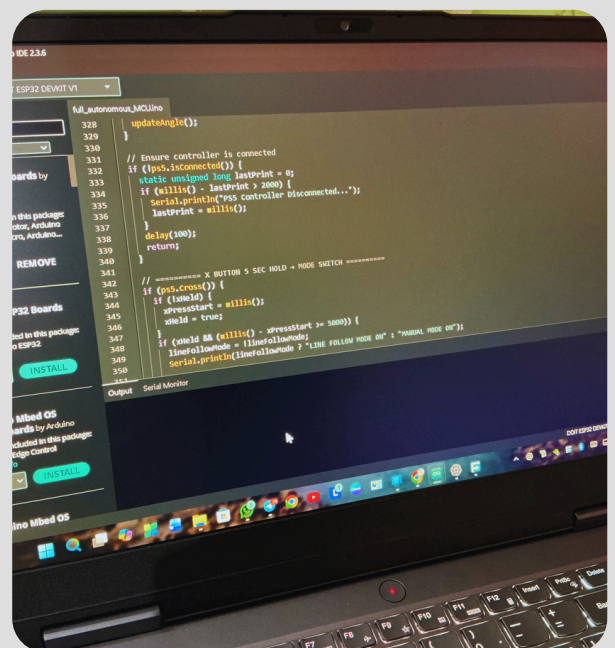
Programming And Controls

BaseBot Software Architecture Overview

The BaseBot software architecture became the digital core of our robot—the system that connected every piece of hardware to the decisions made by the driver. At the centre of this setup was the HEXA Command Hub ESP32, a versatile microcontroller capable of handling motor control, servo movement, Bluetooth communication, and signal processing. Using the Arduino IDE, we uploaded all programs directly to the ESP32, selecting the correct board settings, choosing the right COM port, and ensuring successful compilation before every test run. The Arduino IDE became our programming workspace—where every line of logic, every function, and every movement began.

The software structure communicated through PWM pins and digital signals to the MDD10 motor driver, which controlled the speed and direction of all four DC motors. Meanwhile, dedicated servo pins handled the gripper and arm movements, allowing the robot to interact with scoring elements smoothly. The ESP32 processed real-time controller input from the PS5 gamepad, translating joystick movements into forward motion, turning, lifting, and gripping. This architecture remained simple, clear, and predictable—making it easy for our team to understand how each subsystem responded.

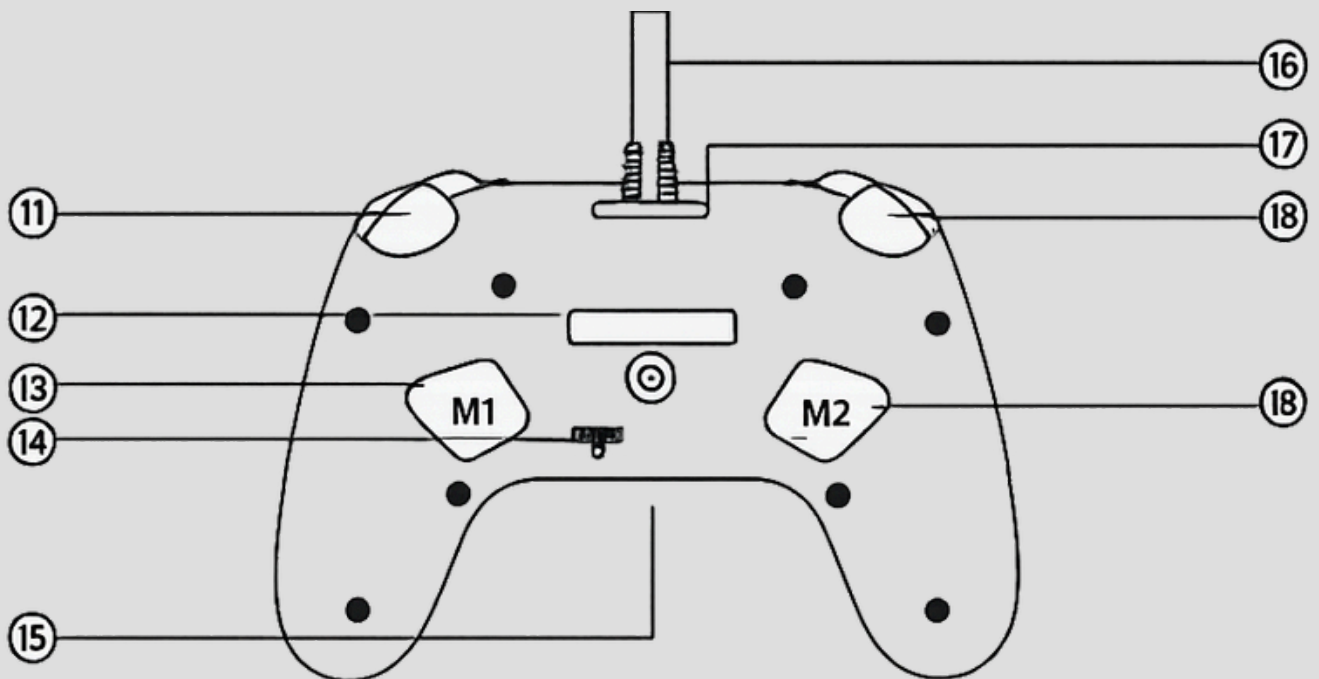
By working within this structure, VoltEdge could focus on refining responsiveness rather than troubleshooting complex communication layers. Every adjustment we made—from speed calibration to turning refinement—began with code written inside Arduino IDE and uploaded to the ESP32. Through this process, the robot developed its digital behaviour, responding cleanly to controls and preparing for advanced autonomous logic. For us, the software architecture was not just code—it was the personality of the machine, built one upload at a time.



Joystick Mapping As Provided By Basebot.

Before creating our own control system, we first learned how the BaseBot behaved using the default joystick mapping provided in the NRL programming guide. The PS5 controller connected to the ESP32 through Bluetooth, and each input was assigned a clear, predefined function. The left joystick controlled forward and backward movement, allowing the robot to accelerate smoothly across the field. The right joystick handled turning, making it possible to rotate or adjust direction with small, precise movements. The basic functional buttons controlled the arm and gripper actions using simple servo commands, giving the robot its ability to lift and release objects.

This mapping gave us a strong foundation for understanding how the robot interpreted driver inputs. Every time we pressed a button or pushed a joystick, the ESP32 translated that signal into motor speed, direction, or servo movement. During early tests, these default mappings helped us observe the robot's natural response—how quickly it turned, how sharply it accelerated, and how the gripper behaved during action. This system became the starting point for VoltEdge, helping us learn the robot's driving personality before we began making custom adjustments tailored to our driver and match strategy.



Program Changes, And Controller Customisation

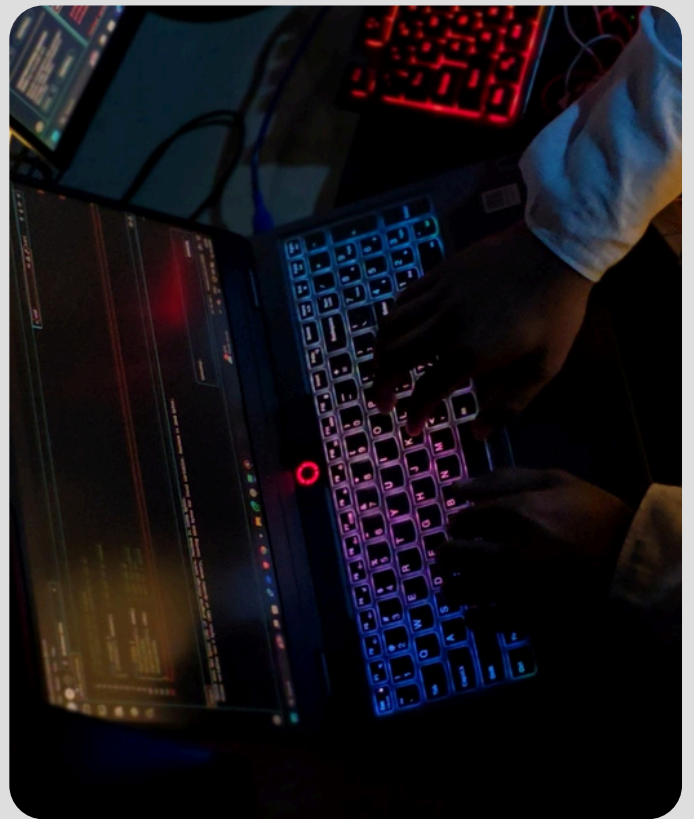


As soon as the BaseBot was assembled and the first drive tests began, VoltEdge realised that the default control layout, although functional, didn't match the pace and precision needed for competition. Under Omer's lead, the team shifted into software refinement, using the Arduino IDE to rebuild the controller mapping into something faster, smoother, and more instinctive for match play. The goal was simple: the driver should never need to think about controls — his hands should respond automatically.

We kept the fundamentals of the BaseBot architecture intact, but customised the mapping to reflect VoltEdge's driving style. The left joystick continued to control forward and backward motion, while the right joystick handled turning and directional adjustments. These remained untouched because they gave Omer the most natural driving flow from the very beginning.

The real transformation came in the mechanism controls. The L2 trigger was assigned as the command to lift the arm up, giving the driver direct finger-based control over elevation without interrupting movement. The R2 trigger became the input for lowering the arm down, allowing quick and controlled repositioning during scoring cycles. To improve object handling speed, the R1 bumper was set to open and close the gripper, giving Omer the ability to react instantly when collecting or releasing charges. This replaced the slower button-based setup and made the robot more efficient during practice runs.

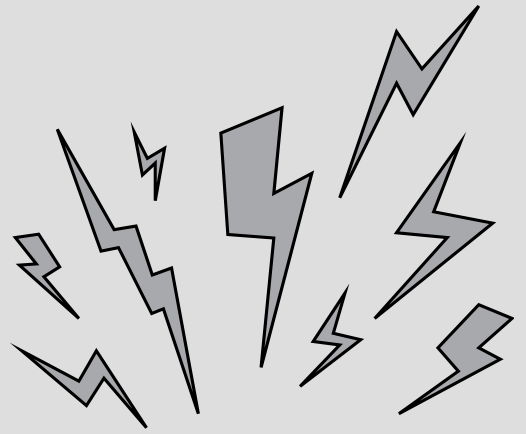
The real transformation came in the mechanism controls. Earlier, the arm moved either fully up or fully down with a single button press, giving the driver no control over stopping midway. After adding PWM and rewriting the movement logic, the arm became far more refined—now it moves smoothly and stops exactly at the point where the driver releases the button. This upgrade alone made handling cycles faster, steadier, and far more predictable during practice.



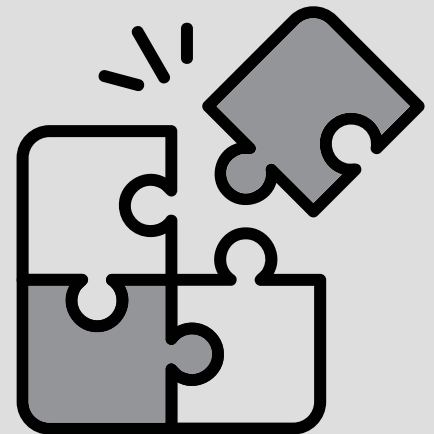
These customisations didn't change the BaseBot — they adapted it to VoltEdge. Through testing, adjustments, and repeated uploads, the robot evolved into a system that moved exactly how the driver thought. The mapping became intuitive, the actions became fluid, and the robot began responding like a natural extension of the driver's hands. By the end of the refinement cycle, EdgeBot no longer felt like a standard build — it felt like our robot, programmed precisely for VoltEdge's rhythm and control.

Troubleshooting And Testing Programs

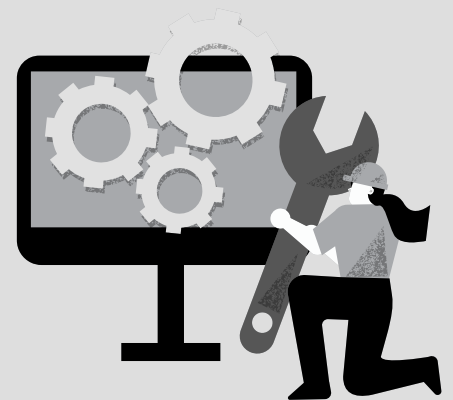
Programming the robot was not a one-time upload — it was a cycle of writing, testing, observing, and correcting. As soon as VoltEdge began experimenting with the BaseBot's controls in the Arduino IDE, small inconsistencies started to appear: a motor turning slightly faster than the opposite one, the arm stopping a bit earlier than expected, or the gripper moving with more force than required. These were not errors — they were cues that the code needed tuning. Under Omer's lead, the team tested drive behaviour repeatedly, adjusting speed mappings, refining turn sensitivity, and recalibrating servo angles until the robot moved smoothly and predictably.



Every evening session became a pattern: upload a version of the code, power on the robot, test the motion on the floor, identify the issue, and head back to the laptop for another correction. Sometimes the robot accelerated too quickly, so the team reduced the response curve of the joysticks. Sometimes the turning radius felt too wide, so the team adjusted the turning factor inside the program. When the gripper required faster reaction time, the servo positions were fine-tuned for cleaner opening and closing. Through this continuous process, EdgeBot gradually became stable, precise, and driver-friendly.



These tests taught the team how every small line of code affected real-world behaviour. Each improvement brought the robot closer to competition readiness. By the end of this phase, the robot was no longer just running code — it was executing movement shaped by VoltEdge's hands, instincts, and problem-solving. The troubleshooting journey became one of the most important learning experiences of the entire build.



Autonomous Skill Challenge

Basebot Autonomous Capabilities

The **Basebot** we received was completely manual.

It had motors, a servo, a chassis, and a controller — but it had no sensors, no autonomous logic, and no pathway to run on its own.

For the Autonomous Skill Challenge, everything had to be built by us from the ground up.

Instead of relying on any built-in features, VoltEdge chose to engineer an autonomous system ourselves. We purchased and integrated our own sensors — a line-following IR array, a gyroscope, and an ultrasonic distance sensor — and transformed a fully manual BaseBot into a robot capable of completing an independent routine.

This meant the BaseBot had zero autonomous capabilities at the start.

Every part of the autonomous system — sensing, logic, reactions, timing — came from VoltEdge's engineering decisions and tests.

The BaseBot did not come with autonomous capabilities.

VoltEdge built them.

Every sensor was chosen by us.

Every behaviour was coded by us.

Every movement in autonomous mode was something we taught the machine ourselves.

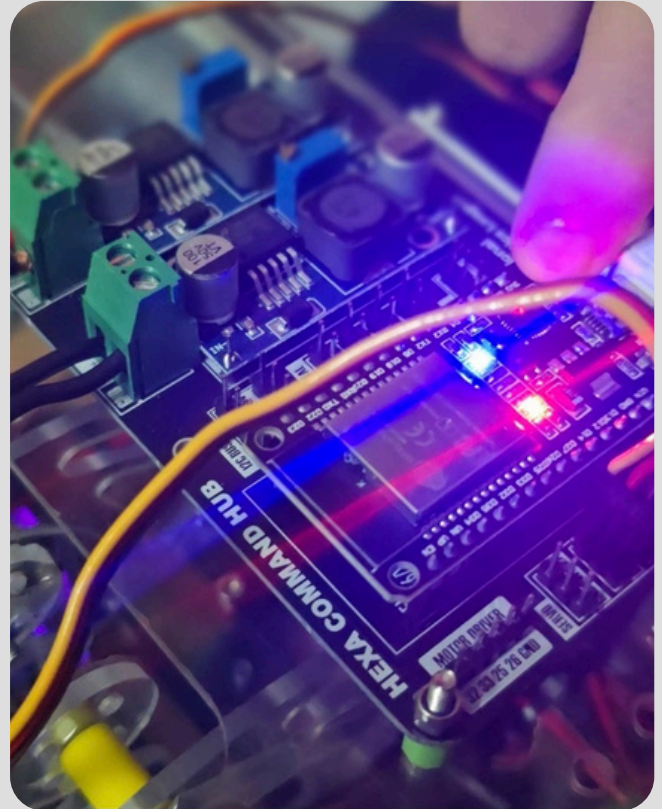
This challenge became one of the most defining moments of our engineering journey — because it wasn't just the robot learning a routine...

it was us learning how to make a robot think.

Our Modifications And Programming Improvements

Since the BaseBot lacked autonomous features and sensors, VoltEdge had to develop the entire system from scratch. We began by equipping the robot with the ability to see, sense, and respond—capabilities it didn't have initially. We acquired a 5-sensor IR line array, gyroscope, and ultrasonic sensor, and integrated these components into a robot meant for manual control.

The IR array became the cornerstone of our autonomous navigation. Positioned at the front, it detected light and dark surfaces, enabling the robot to recognize when it was on or off the line. This required meticulous calibration and tailored correction logic to ensure smooth adjustments without jolting or overshooting turns.



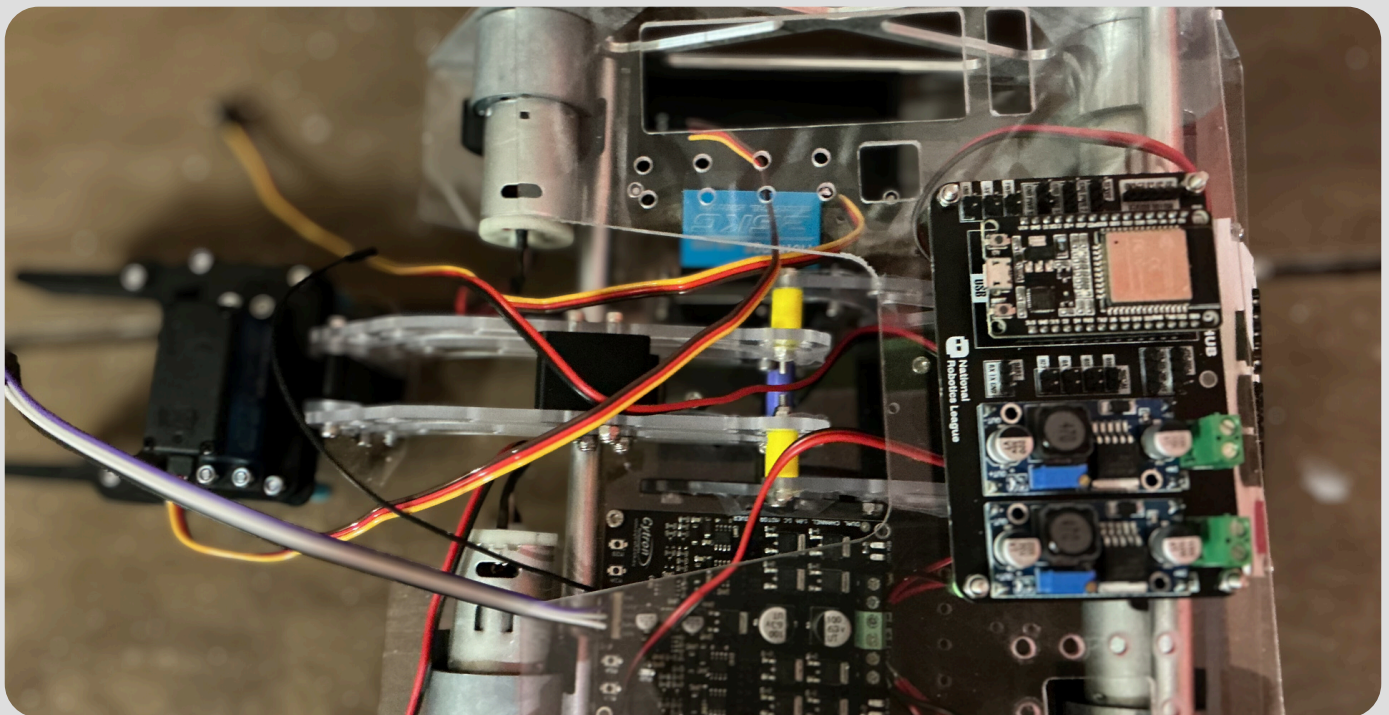
We added a gyroscope to stabilize the robot's orientation. Before this, the robot drifted slightly during tests and struggled to maintain consistent angles. With the gyroscope, it could hold its direction, correct rotational errors, and execute accurate turns—essential for completing an autonomous routine.

We also attached an ultrasonic sensor to help the robot detect charging stations from a safe distance. It was programmed to slow down when the ultrasonic reading exceeded a certain threshold and to stop precisely for alignment. This minimized collision risks and ensured a smooth, predictable approach.



All of these abilities were built through programming improvements. We wrote custom logic inside the Arduino IDE to:

- Read all five IR sensors and determine correction directions
- Use gyro feedback to stabilize turns and straight paths
- Process ultrasonic distance data for slowing and stopping
- Blend timing, sensor readings, and motor responses into one routine



To meet challenge rules, we added a deliberate activation method: holding the X button for five seconds triggers autonomous mode. This ensured safety and avoided accidental starts.

Through repeated uploads, tests, and refinements, the BaseBot evolved from a fully manual machine into a robot capable of running an entire routine independently. Every correction, every angle, every movement in autonomous mode was the result of VoltEdge's own design — a custom-built system shaped by our logic, sensors, and persistence.